DATA SCIENCE

Machine Learning Starter Pack

7 algorithms that cover most work, the metrics that matter, and a 60-day plan from zero.

The 7 algorithms

Algorithm	Type	Typical use
Linear Regression	Regression	Forecast sales, price
Logistic Regression	Classification	Churn yes/no, loan default
Decision Tree	Both	Explainable rules
Random Forest	Both	Strong baseline on tabular data
Gradient Boosting (XGBoost / LightGBM)	Both	Best accuracy on tabular data
K-Means	Clustering	Customer segments
KNN	Both	Simple similarity, recommendations

Metrics

Metric	When
Accuracy	Only when classes are balanced
Precision	False positives are costly (spam filter)
Recall	False negatives are costly (fraud, disease)
F1	Balance of precision and recall
ROC-AUC	Ranking quality across thresholds
RMSE / MAE	Regression error

"Your model is 99% accurate. Why is that bad?" If only 1% of transactions are fraud, a model that predicts "not fraud" every time is 99% accurate and catches nothing. Check recall, precision and the confusion matrix. Fix with resampling, class weights or a better threshold.

Basics every interviewer checks

- **Overfitting:** great on train, poor on test. Fix with more data, simpler model, regularisation, cross-validation.
- **Train / validation / test split** and why you never tune on test.
- **Data leakage:** a feature that would not be known at prediction time.
- **Bias vs variance** trade-off.
- **Feature scaling** matters for KNN, K-Means, logistic regression.

```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.2, stratify=y, random_state=42)
model = RandomForestClassifier(class_weight="balanced").fit(X_tr, y_tr)
print(classification_report(y_te, model.predict(X_te)))
```

60-day plan from zero

Days	Focus
1-10	Python and Pandas basics
11-20	Statistics: distributions, hypothesis testing, regression
21-35	Scikit-learn: the 7 algorithms, metrics, cross-validation
36-45	Feature engineering, imbalanced data, tuning
46-60	Two end-to-end projects with a README and a short demo

5 projects that get shortlisted

1. Customer churn prediction with business impact in ₹.
2. Sales forecasting for the next 3 months.
3. Customer segmentation with K-Means and named segments.
4. Credit risk scoring with an explainability section.
5. A/B test analysis with a clear recommendation.

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